

## Special right triangles



1 In a  $30^\circ - 60^\circ - 90^\circ$  triangle, identify which leg is shorter.

2 Find the exact value of the following:

a  $\sin 60^\circ$

b  $\cos 60^\circ$

c  $\tan 30^\circ$

d  $\sin 45^\circ$

3 Consider the following equilateral triangle.  
Find the exact values of the following:

a Perpendicular height of the triangle

b Size of angle  $x$

c

i  $\sin 60^\circ$

ii  $\cos 60^\circ$

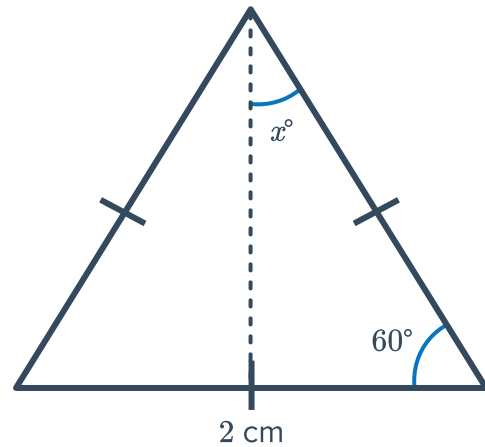
iii  $\tan 60^\circ$

d

i  $\sin x$

ii  $\cos x$

iii  $\tan x$



## Exact value trigonometric ratios

4 Consider  $\sin \theta = \frac{\sqrt{3}}{2}$ .

a Find the value of  $\theta$ , given that  $0^\circ \leq \theta \leq 90^\circ$ .

b Now, find the value of  $\cos \theta$ .

5 Consider  $\sin \theta = \frac{1}{\sqrt{2}}$ .

a Find the value of  $\theta$ , given that  $0^\circ \leq \theta \leq 90^\circ$ .

b Now, find the exact value of  $\tan \theta$ .

6 Consider  $\cos \theta = \frac{\sqrt{3}}{2}$ .



a Find the value of  $\theta$ , given that  $0^\circ \leq \theta \leq 90^\circ$ .

b Now, find the value of  $\sin \theta$ .

7 Consider  $\cos \theta = \frac{1}{2}$ .

a Find the value of  $\theta$ , given that  $0^\circ \leq \theta \leq 90^\circ$ .

b Now, find the value of  $\tan \theta$ , correct to one decimal place.

8 Consider  $\tan \theta = 1$ .

a Find the value of  $\theta$ , given that  $0^\circ \leq \theta \leq 90^\circ$ .

b Now, find the value of  $\sin \theta$ , correct to two decimal places.

9 Consider  $\tan \theta = \frac{1}{\sqrt{3}}$ .

a Find the value of  $\theta$ , given that  $0^\circ \leq \theta \leq 90^\circ$ .

b Now, find the value of  $\sin \theta$ .

10 Consider  $\tan \theta = \frac{1}{\sqrt{3}}$ .

a Find the value of  $\theta$ , given that  $0^\circ \leq \theta \leq 90^\circ$ .

b Now, find the value of  $\cos \theta$ , correct to one decimal place.

11 Consider  $\cos \theta = \frac{\sqrt{3}}{2}$ .

a Find the value of  $\theta$ , given that  $0^\circ \leq \theta \leq 90^\circ$ .

b Now, find the value of  $\tan \theta$ , correct to two decimal places.

12 Consider  $\sin \theta = \frac{\sqrt{3}}{2}$ .

a Find the value of  $\theta$ , given that  $0^\circ \leq \theta \leq 90^\circ$ .

b Now, find the value of  $\tan \theta$ , correct to two decimal places.

13 Consider  $\tan \theta = \sqrt{3}$ .



- a Find the value of  $\theta$ , given that  $0^\circ \leq \theta \leq 90^\circ$ .
- b Now, find the value of  $\sin 60^\circ$ , correct to one decimal place.

14 For each of the following, find the value of  $\theta$  given that  $0^\circ \leq \theta \leq 90^\circ$ :

a  $\cos \theta = \frac{1}{\sqrt{2}}$

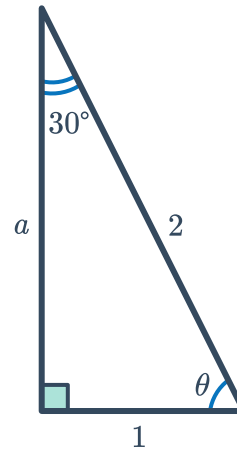
b  $\tan \theta = \sqrt{3}$

c  $\sin \theta = \frac{\sqrt{3}}{2}$

d  $\tan \theta = \frac{1}{\sqrt{3}}$

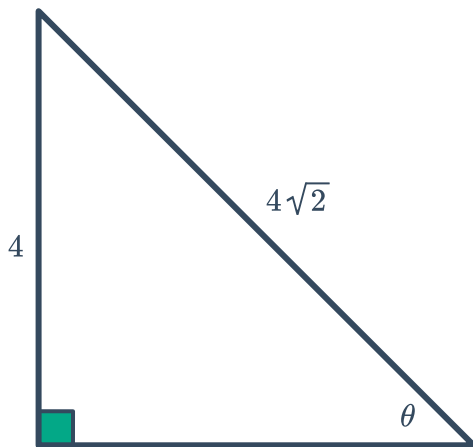
15 Consider the following triangle:

- a Find the exact value of  $a$ .
- b Find the value of  $\theta$ .

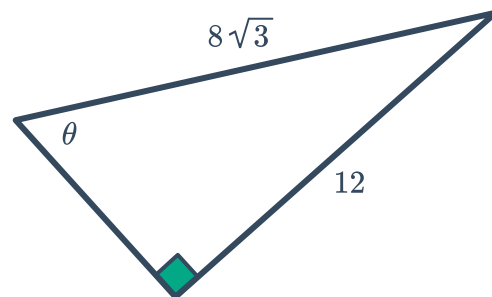


16 Use exact values to solve for  $\theta$  in the following triangles:

a



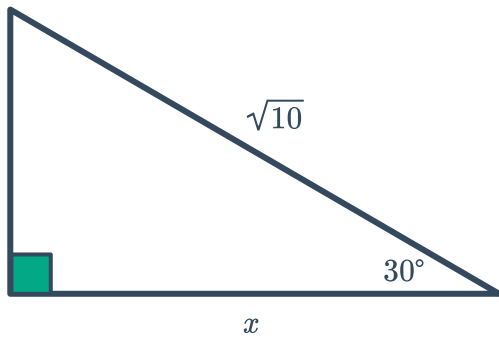
b



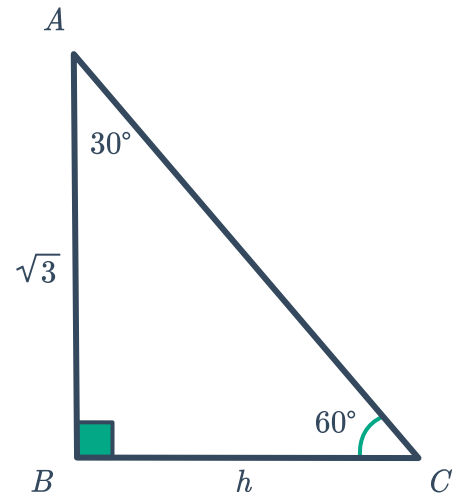
17 Find the exact value of the variable(s) in the following triangles:



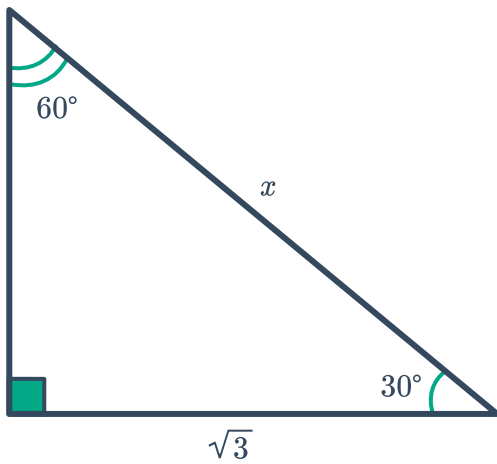
a



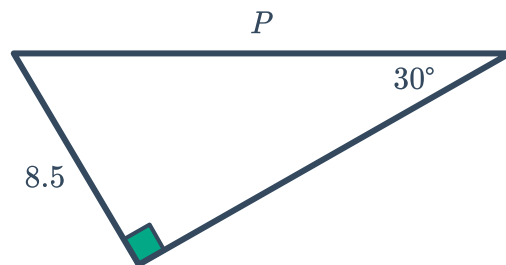
b



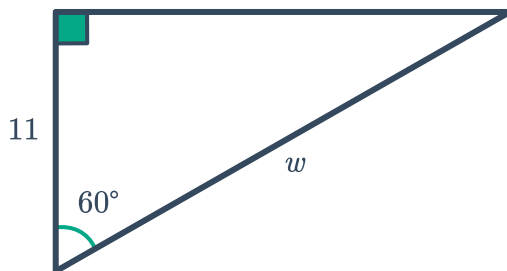
c



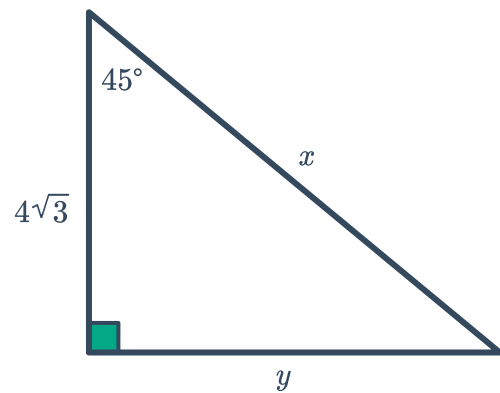
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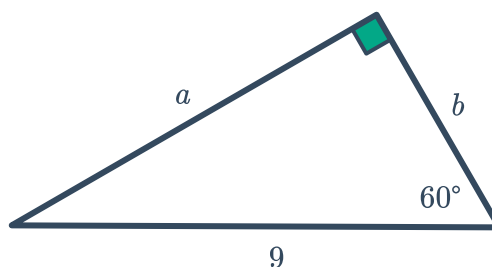
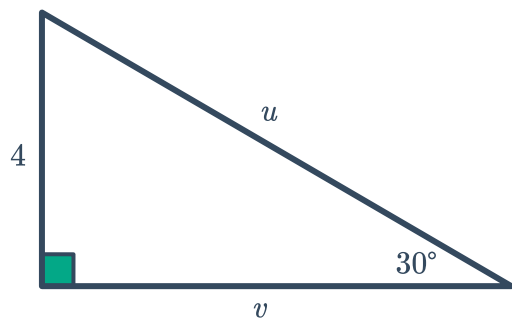
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f

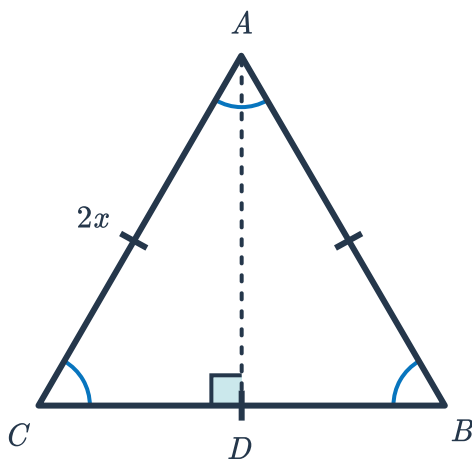


g



## Applications

- 18  $\triangle ABC$  is an equilateral triangle with side lengths of  $2x$  units. An altitude is drawn from  $A$  to meet side  $\overline{BC}$  at  $D$ . This information is shown in the diagram below.



- Find the  $m\angle B$ .
- Given that  $\overline{AD}$  bisects  $\angle BAC$ , find the measure of  $\angle BAD$ .
- Given that  $\overline{AD}$  bisects  $\overline{BC}$ , find the length of  $\overline{BD}$ .
- Determine the length of the altitude  $\overline{AD}$  in terms of  $x$ .
- State whether the following statements are true or false. If false, correct the statement so it is true.
  - The longer leg of a  $30^\circ$  -  $60^\circ$  -  $90^\circ$  triangle is twice the length of the shorter leg.
  - The shorter leg of a  $30^\circ$  -  $60^\circ$  -  $90^\circ$  triangle is opposite the angle measuring  $30^\circ$ .
  - The hypotenuse of a  $30^\circ$  -  $60^\circ$  -  $90^\circ$  triangle is  $\sqrt{3}$  times the length of the longer leg.
  - The ratio of sides in a  $30^\circ$  -  $60^\circ$  -  $90^\circ$  triangle is  $1 : \sqrt{3} : 2$ .



19 Consider the figure shown:

a Find the exact length of the following:

i  $p$     ii  $q$     iii  $r$     iv  $s$

b Now, find the exact perimeter of the quadrilateral.

